

Operational semantics for Prolog with Cut in Rocq and its application to determinacy analysis

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Abstract

We mechanize two operational semantics for PROLOG with cut and prove them equivalent, in the Rocq prover. The first semantics closely models the ELPI’s implementation, it is based on a stack of choice points and is well suited for studying optimizations employed by PROLOG abstract machines. The second semantics is instead based on and-or trees, in which the branches pruned by the cut operator are made explicit.

We use the tree-based semantics (1) to prove properties about the effect of the cut operator in the evaluation of choice points, since the rules from classical logic do not apply, and (2) to reason about the determinacy analysis introduced in Elpi version 3.0, which statically identifies computations that produce at most one result.

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1 Introduction

ELPI is a dialect of λ PROLOG (see [14, 15, 7, 12]) used as an extension language for the ROCQ prover (formerly the COQ proof assistant). ELPI has become an important infrastructure component: several projects and libraries depend on it [13, 3, 4, 19, 8, 9]. Examples include the Hierarchy-Builder library-structuring tool [5] and Derive [17, 18, 11], a program-and-proof synthesis framework with industrial applications at SkyLabs AI.

Starting with version 3, ELPI is equipped with a static analysis for determinacy [10] to help users tame backtracking. ROCQ users are familiar with functional programming but not necessarily with logic programming, in particular uncontrolled backtracking is a common source of inefficiency and makes debugging harder. The determinacy checker identifies predicates that behave like functions, i.e., predicates that commit to their first solution and leave no *choice points* (places where backtracking could resume).

This paper reports our first steps towards a mechanization, in the ROCQ prover, of the determinacy analysis from [10]. We focus on the control operator *cut*, which is useful to restrict backtracking but makes the semantics depart from a pure logical reading.

We formalize two operational semantics for PROLOG with cut. The first is a stack-based semantics that closely models ELPI’s implementation and is similar to the semantics mechanized by Pusch in ISABELLE/HOL [16] and to the model of Debray and Mishra [6, Sec. 4.3]. This stack-based semantics is a good starting point to study further optimizations used by standard PROLOG abstract machines [20, 1], but it makes reasoning about the scope of cut difficult. To address this limitation we introduce a tree-based semantics in which the branches pruned by *cut* are explicit and we prove the two semantics equivalent. Using the



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Inductive P := IP of nat. Inductive D := ID of nat. Inductive V := IV of nat.

Inductive Tm :=
  | Tm_P of P
  | Tm_D of D
  | Tm_V of V
  | Tm_App of Tm & Tm.

Record Unif := {
  unify : Tm → Tm → Σ → option Σ;
  matching : Tm → Tm → Σ → option Σ;
}.

```

■ Figure 1 Tm types

■ Figure 2 Unif type

tree-based semantics, we then prove some properties about the impact of evaluating the *cut* in query disjunctions. Unlike in classical logic, $q_1 \vee q_2$ is not equivalent to $q_2 \vee q_1$, since q_2 may be cut-away by q_1 . Moreover, we show that if every rule of a predicate passes the determinacy analysis, then a call to that predicate does not leave any choice points.

2 Preliminaries: syntax and informal semantics

Before going to the two semantics, we show the piece of data structure that are shared by the them. The smallest unit of code that we can use in the language is an atom. The atom inductive (see Type 1) is either a cut or a call. A call carries a term (see Figure 1). A term (Tm) is either a predicate, a datum, a variable or the binary application of two terms.

```

Inductive A := cut | call : Tm → A. (1)
Record R := mkR { head : Tm; premises : list A }. (2)
Record P := { rules : seq R; sig : sigT }. (3)
Definition Σ := {fmap V → Tm}. (4)
Definition bc : P →  $\mathcal{F}_V$  → Tm → Σ →  $\mathcal{F}_V$  * seq (Σ * seq A) := (5)

```

A rule (see Type 2) has a head of type term and a list of premises, that are a list of atoms. A program (see Type 3) is made by its rules and the signatures of predicates, **sigT** is a mapping from predicates to their signatures, signatures represent the definition of type from the simply typed lambda calculus, i.e. it is either a base type or an arrow for term application. We decorate arrows to know the mode (input or output) of the application argument.

A substitution (see Type 4) is a mapping from variables to terms. It is the output of a successful query. The empty substitution is noted ε .

The backchain function (bc, see Type 5) takes: a program P ; a set fv of variables ($\mathcal{F}_V$); a query q ; and the substitution σ . A backchain is performed when the current atom is a **call**. In this case, it start by recursively *dereferencing* the variables in σ , i.e. replacing all the variables in the query by their term mapping in σ .

TODO: free variables and program freshness

Then it takes the head of the term, which should be rigid, i.e. a predicate p , and look for the signature s of p in the program. Then, for each rule r in P , it unifies the head of r with q . Unification is possible thanks to the unificator U , whose type is shown in Figure 2. U explains how to unify terms up to standard unification (for output terms) or matching (for input terms). The function *unif* (resp. *matching*) unifies (resp. matches) the two terms under σ , it return an extension of σ if the two terms can be unified (resp. matched), and None otherwise. The result of a backchain is made by filtering the rules in P that can be used on the given query. In particular, for each rule filtered rules r , it returns a pair containing the extension of σ making the head of r and q unify and the list of premises of r .

```

f 1 2.    f 2 3.    r 0 0.    r 0 1.    r 0 2.
g X Z :- r X Y, f Y Z, !. % r1
g X Z :- f X Y, f Y Z.    % r2

```

Inductive tree :=
 | KO | OK | TA of A
 | Or of option tree & Σ & tree
 | And of tree & seq A & tree.

■ Figure 3 Small ELPI program example

■ Figure 4 Tree inductive

2.1 The cut operator

ELPI is a dialect of λ PROLOG with the hard cut operator and, in the language we have shown before, *cut* is an atom. The semantics of the cut operator adopted in the ELPI language corresponds to the *hard cut* operator of standard SWI-PROLOG. This operator has two primary purposes. First, it eliminates all alternatives that are created either simultaneously with, or after, the introduction of the cut into the execution state.

The ELPI program we will use as an example in the following sections is shown in Figure 3.

To illustrate the high-level description of *cut*, consider the program P shown in Figure 3 and the query $q = g \ 0 \ R$. Both rules for g can be used on the query q . Backchaining q in P from the empty (ε) substitution returns the list $[(\sigma_1, [r \ X \ Y, f \ Y \ Z, !]), (\sigma_2, [f \ X \ Y, f \ Y \ Z])]$ with σ_1 and σ_2 both equal to $\{X \mapsto 0, Z \mapsto R\}$. The two elements of the list are generated respectively from the rules r_1 and r_2 .

The next step of the interpretation corresponds to the backchaining of $r \ X \ Y$ from σ_1 . This returns $[(\sigma_3, []), (\sigma_4, []), (\sigma_5, [])]$ with $\sigma_3 := \sigma_1 + \{Y \mapsto 0\}$, $\sigma_4 := \sigma_1 + \{Y \mapsto 1\}$ and $\sigma_5 := \sigma_1 + \{Y \mapsto 2\}$. The three lists of premises are empty since all rules for r have no premises.

Backchaining $f \ Y \ Z$ in σ_3 gives no solution, therefore, by backtracking, we backchain $f \ Y \ Z$ in σ_4 . This gives a solution, namely $\sigma_6 := \sigma_4 + \{R \mapsto 2\}$.

We now reach the cut operator and can see its double side behavior. In the current execution state, we have still two unexplored alternatives: we can backtrack on the third solution for r that assign Y to 2 and on the rules r_2 . We can note, however, the first alternative have been generated after the cut introduction, and the second alternative have been generated at the same moment of the cut. Therefore both are cut away leaving no choice point. The final complete solution is: $\{X \mapsto 0, Z \mapsto R, Y \mapsto 1, R \mapsto 2\}$, where X , Y and Z can be ignored since they are the result internal unification.

We propose two formal, operational semantics for a logic program with cut. The two semantics are based on different syntaxes, the first syntax (called *tree*) exploits a tree-like structure and is ideal both to have a graphical view of its evolution while the tree is being interpreted and to prove lemmas over it. The second syntax, called *elpi*, is the ELPI's syntax and has the advantage of reducing the computational cost of cutting and backtracking alternatives by using shared pointers. We aim to prove the equivalence of the two semantics together with some interesting lemmas of the cut behavior.

Notations used in the paper In the following section we note $\mathbb{B}$ for the boolean type, and $\top$ and $\perp$ for **true** and **false**. The option instances are noted $\bullet t$ for *Some t* and $\square$ for *None*.

se metti $r1 = g \ A \ B :- f \ A \ B$. allora $g \ e \ f$ sono funzioni, e puoi spiegare anche l'idea del detcheck qui

spiegare notazione for list

3 And-Or Tree semantics

The tree state The tree inductive, shown in Figure 4. We distinguish 5 main cases: KO , OK , and are special meta-symbols representing, respectively, the basic failed and a successful trees. These symbols are considered meta because they are internal intermediate symbols

```

Fixpoint get_end s A :  $\Sigma$  * tree :=
  match A with
  | TA _ | KO | OK  $\Rightarrow$  (s, A)
  | Or  $\square$  s1 B  $\Rightarrow$  get_end s1 B
  | Or •A _  $\Rightarrow$  get_end s A
  | And A _ B  $\Rightarrow$ 
    let (s', pA) := get_end s A in
    if pA == OK then get_end s' B
    else (s', pA)
  end.

Definition get_subst s A := (get_end s A).1.
Definition path_end A := (get_end  $\varepsilon$  A).2.
Definition success A := path_end A == OK.
Definition failed A := path_end A == KO.
Definition path_atom A :=
  if path_end A is TA _ then  $\top$ 
  else  $\perp$ .

```

(a) Definition of *get_end*(b) Functions from *get_end*

■ **Figure 5** *get_end* and derives functions

119 used to give structure to the tree.

120 The *TA* constructor (acronym for tree-atom) is the constructor of atoms in the tree. They
 121 are either the cut operator or a call, in these cases, the interpreter should evaluate the atom
 122 by either cutting the subtrees in the scope of the cut or backchaining the call in the program.

123 The two recursive cases of a tree are the *Or* and *And* nodes. The *Or* node $A \vee B_\sigma$ denotes
 124 a disjunction between two trees *A* and *B*. The first branch is optional, if absent it represents
 125 a dead tree, i.e. a tree that has been entirely explored. The second branch is annotated
 126 with a suspended substitution σ so that, upon backtracking to *B*, σ is used as the initial
 127 substitution for the execution of *B*.

128 The *And* node $A \wedge_{B_0} B$ represents a conjunction of two trees *A* and *B*. We call B_0 the
 129 reset point for *B*; it is used to restore the state of *B* to its initial form if a backtracking
 130 operation occurs on *A*. An example of this behavior is shown in Section 3.2

131 A graphical representation of a tree is shown in Figure 6a. To make the graph more
 132 compact, the *And* and *Or* nodes are n-ary rather than binary, and the children of these
 133 nodes associate to the right. The *KO* node acts as the neutral elements in the *Or* list, while
 134 *OK* is the neutral element of the *And* list.

135 *The tree interpreter* The interpretation of a tree is performed by two main routines: *step*
 136 and *next_alt* that traverse the tree depth-first, left-to-right. Then, then *runT* inductive
 137 makes the transitive closure of *step* and *next_alt*: it iterates the calls to its these
 138 auxiliary functions as much as needed. In Types 7–9 we give the types contrats of these
 139 symbols where $\mathcal{F}_v$ is a set of variable names.

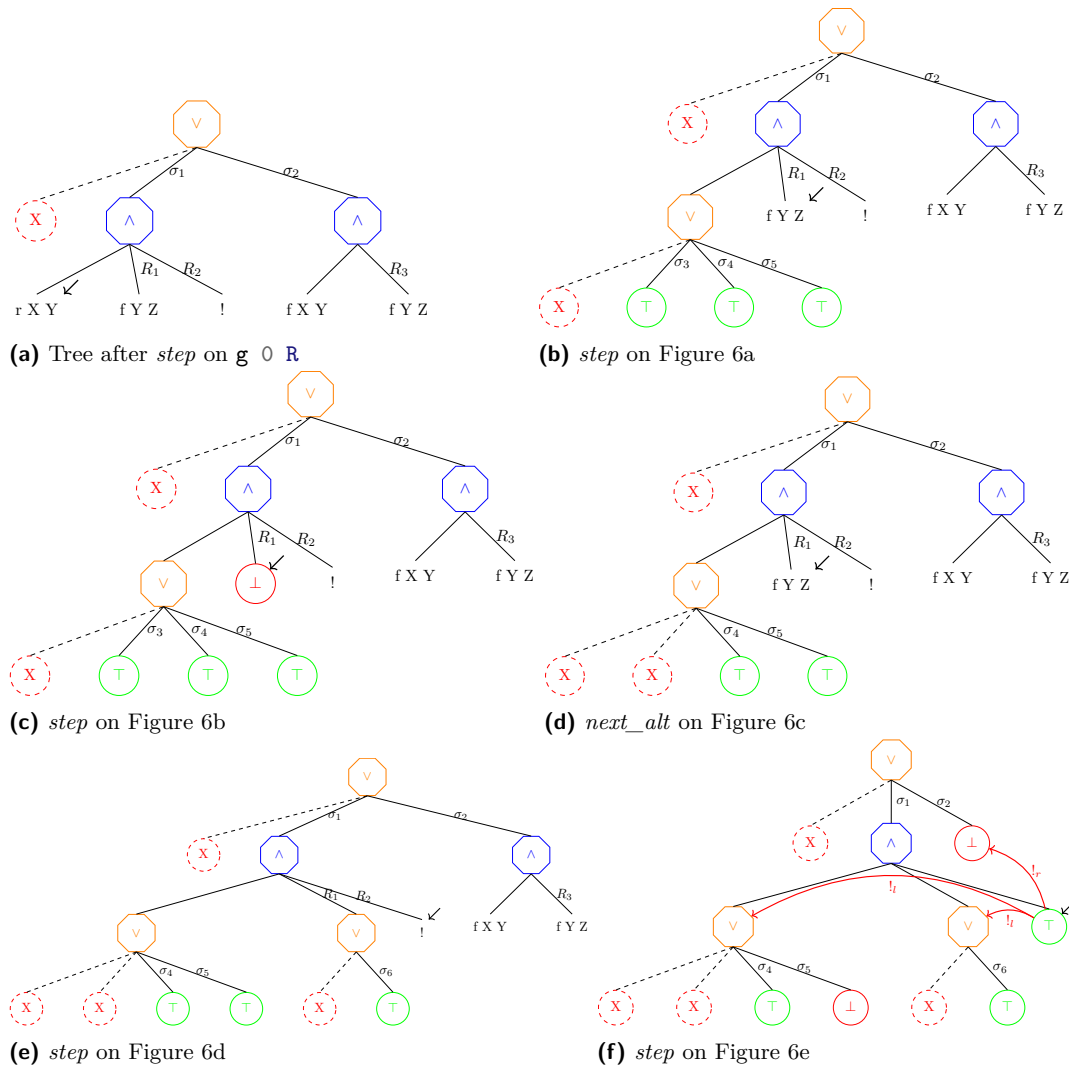
140 **Inductive** tag := Expanded | CutBrothers | Failed | Success. (6)

141 **Definition** step : $\mathbb{P} \rightarrow \mathcal{F}_v \rightarrow \Sigma \rightarrow \text{tree} \rightarrow (\mathcal{F}_v * \text{tag} * \text{tree}) :=$ (7)

142 **Definition** next_alt : $\mathbb{B} \rightarrow \text{tree} \rightarrow \text{option tree} :=$ (8)

143 **Inductive** runT (p : $\mathbb{P}$): $\mathcal{F}_v \rightarrow \Sigma \rightarrow \text{tree}$
 $\rightarrow \text{option } (\Sigma * \text{option tree}) \rightarrow \mathbb{B} \rightarrow \mathcal{F}_v \rightarrow \text{Prop} :=$ (9)

144 The tree interpreter, as in prolog, explores the tree in DFS strategy, to discover the
 145 substitution and the leaf of the tree that should be interpreted. These two pieces of
 146 information can be retrieved for a substitution *s* and a tree *A* thanks to the *get_end* routine,
 147 shown in Figure 5a If the tree is already a leaf, *get_end* returns its inputs. Otherwise, if the
 148 tree is a disjunction, the path continues on the left subtree, if it exists, otherwise *get_end*
 149 continues on the rhs using the substitution stored in the *Or* node. In the case of a conjunction,
 150 if *get_end* of the lhs returns the tree *OK*, then the *get_end* continues in the rhs, otherwise



■ **Figure 6** Some tree representations¹

we return the result of the lhs. We derive some useful definition from *get_end*. They are listed in Figure 5b.

We have pointed by a black arrow the *path_end* of all the trees in Figure 6.

3.1 The *step* procedure

The *step* procedure takes as input a program, a set of variables, a substitution, and a tree, and returns an updated set of variables, a *tag*, and an updated tree.

The set of variables v given in input to *step* are used and updated when a backchaining operation takes place. A call to *step* produces a good result if v contains all the variables in the tree. This guarantees that any call to *bc* returns a list of atoms whose variables are disjoint from the variables of the current working tree.

not true in figure:fix + also fix in Section 2.1

¹ The substitution $\sigma_1 \dots \sigma_6$ are from Section 2.1, R_1, R_2 and R_3 are the reset points and correspond

The *tag* (see Type 6) indicates the kind of an internal tree step: **CutBrothers** denotes the interpretation of a superficial cut, i.e., a cut whose parent nodes are all *And*-nodes. **Expanded** denotes the interpretation of non-superficial cuts or predicate calls. **Failure** and **Success** are returned for, respectively, *failed* and *success* trees.

The step procedure is intended to evaluate atoms, that is, it leaves unchanged the tree iff its *path_end* is not an atom, i.e. if it is a success- or a failed-tree.

Lemma succ_step_iff: $\forall p \ v \ s \ A, \text{success } A \leftrightarrow \text{step } p \ v \ s \ A = (v, \text{Success}, A).$ (1)

Lemma fail_step_iff: $\forall p \ v \ s \ A, \text{failed } A \leftrightarrow \text{step } p \ v \ s \ A = (v, \text{Failed}, A).$ (2)

The interpretation of a call to a term *c* starts by calling the *bc* function on *c*. The result of the backchain is a list *l*. If *l* is empty then the current call is replaced by the *KO* node, otherwise it is replaced by a right skewed tree made by *Or* inner-nodes and each leave correspond to a list of atom *e* in *l*. If *e* is empty then it is mapped to the *OK* tree, otherwise it is mapped to a right-skewed tree made by *And* inner-nodes.

sottolineare
sostituzioni

In Figure 6a, we provide a clear example of a call to *step* on the query $q \ 0 \ R$ on the program *P* in Figure 3. Let *c*₀, *c*₁ and *c*₂ be the children of the root. *c*₀, by construction is the *None* tree, while *c*₁ and *c*₂ correspond respectively to the premises of rules *r*₁ and *r*₂ in *P*. We need the first *None* subtree so that we can attach a substitution to *c*₁. In particular, we note that the *c*₁ (resp. *c*₂) are related to the substitution σ_1 (resp. σ_2). The value of σ_1 and σ_2 are the same as in Section 2.1. The reset points in the tree are marked with the *R* symbol. $R_1 := [f \ Y \ Z, !]$, $R_2 := [!]$ and $R_3 := f \ Y \ Z$, they are essentially the list of atoms that allowed to generate the current subtree. Other example of call to the *step* procedure triggering a backchain are Figures 6b, 6c, and 6e.

The interpretation of a cut has three main impacts: at first the cut is replaced by the *OK* node, then some special subtrees, in the scope of the *Cut*, are cut away: in particular we need to soft-kill the left-siblings of the *Cut* and hard-kill the right-uncles of the *Cut*.

► **Definition 1** (Left-siblings (resp. right-sibling) and Right-uncles). *Given a node A, the left-siblings (resp. right-sibling) of A are the list of subtrees sharing the same parent of A and that appear on its left (resp. right). The right-uncles of A are the list of right-sibling of the father of A.*

► **Definition 2** (Hard-kill, $!_r$ and Soft-kill, $!_l$). *Given a tree t, hard-kill replaces the given subtree with the KO node. If t is a success-tree, then soft-kill replaces with KO all subtrees that are not part of the path in t leading to the OK node.*

An example of the impact of the cut is show in Figure 6f. In Figure 6e, we see that the *path_end* is a cut. The cut is replaced by the *OK* node and it need to cut-away some alternatives in the tree. The red arrows indicate the scope of the cut, they are labeled with $!_r$ and $!_l$ to explain which kind of cut has been performed on each pointed subtree. We note that, after the evaluation of a cut, the path leading to the cut node is not affected by $!_l$, the success under the substitution σ_4 is kept, while the success under the substitution σ_5 is replaced by the *KO* node. If we refer to the paragraph in Section 2.1, we are discarding the third rule of the *r* predicate.

respectively to the list of atoms $[f \ Y \ Z, !]$, $[!]$ and $[f \ Y \ Z]$

3.2 The *next_alt* procedure

The *step* procedure alone is not sufficient to reproduce entirely the behavior of the full expected prolog interpreter. As proved in Lemma 1 and Lemma 2, *step* does not make any progression on the tree being interpreted. To solve this problem, we need a routine capable of backtracking in case of failures. This routine return the backtracked tree if it exists, *None* otherwise.

In Figure 6c, we are facing a failure: the interpreter has evaluated the atom $\mathbf{f} \ X \ Y$ under the substitution σ_3 which assigns Y to 0, but, in Figure 3, we have no rule satisfying this query. The role of *next_alt*, in this example, is to kill the path leading to this failure, while resetting the failure to its original shape. We see how the reset point plays a major role in this case. At first we are killing the *OK* node under the substitution σ_3 , since this tree have been entirely explored and led to no solution, then we replace the *KO* node with the information stored in R_1 . That is, thanks to R_1 , we are restoring the $\mathbf{f} \ Y \ Z$ call, so that we are able to evaluate this node under the substitution σ_4 . More concretely, we have uncessesfully tried rule $\mathbf{r} \ 0 \ 0$, therefore, we need to continue by trying rule $\mathbf{r} \ 0 \ 1$.

Furthermore, *next_alt* accomplishes to another important task. Its behavior changes depending to the value of the boolean it receives in entry: *next_alt false A* aims to recursively cancel all the failures (if any) in A , whereas, *next_alt true A* aims to cancel the first success in A . This second behavior is helpful since, we need to progress the tree in case of success. For example, consider the tree in Figure 6f, it contains a success, mapping namely R to 2. The interpreter should return this sucess, but, due to the “non-deterministic” behavior of the prolog system, we also need to return all the other solution that could be available by evaluating the rest of the tree. After the replacement of the *path_end* in Figure 6f with *KO*, we remain with a tree having all failing path. The *next_alt* procedure keeps removing them until discovering that the tree have no worth-to explore paths, therefore it will return *None*.

Some interesting property of *next_alt* are shown below and allow to see how *next_alt* complements *step* wrt failed, success and *path_atom* trees.

$$\text{Lemma path_atom_next_alt_id: } \forall b \ A, \text{ path_atom } A \rightarrow \text{next_alt } b \ A = \bullet A. \quad (3)$$

$$\text{Lemma next_alt_failedF: } \forall b \ A \ A', \text{ next_alt } b \ A = \bullet A' \rightarrow \text{failed } A' = \perp. \quad (4)$$

$$\text{Lemma next_altFN_fail: } \forall A, \text{ next_alt } \perp \ A = \square \rightarrow \text{failed } A. \quad (5)$$

3.3 The *runT* inductive

The inductive *runT* is modeled as a function taking 4 inputs and returning 3 outputs. The inputs are the program, a set of variables v_0 , an initial substitution σ_0 , and a tree t_0 . The outputs are an optional object r containing a substitution σ_1 and an optional tree t_1 , and a boolean b and a set of updated variables v_1 . The very interesting piece of information is the result r , if *None*, it tells that the interpretation failed, otherwise, σ_1 represents the most-general unificator that makes the execution of the tree t_0 under the initial substitution σ_0 succeeds: σ_1 is an extension of σ_0 . The tree t_1 is the updated tree containing the alternatives that have not yet been explored, if these alternatives do not exists, t_1 is *None*. The output b and v_1 are there to tell respectively if during the execution a superficial cut has been evaluated, and what is the updated set of variables after the interpretation.

The procedure *runT* is based on four main derivation rules, shown in Figure 7. (STOPT) If the *path_end* of the tree t_0 is a success, the substitution σ_1 is retrived in t_0 from σ_0 and the input tree is cleaned of its successful path. (STEPT) If the *path_end* of t_0 is an atom, then *step* is invoked to evaluate t_0 , and *runT* is recursively called on the new tree. (BACKT)

246 If the *path_end* of t_0 is a failure, *next_alt* is called to clear the failed path; if the resulting
 247 cleaned tree t_1 exists, *runT* is recursively called on it. Finally, (FAILT) takes into account
 248 tree with no solutions.

249 **Lemma** *runT_det*: $\forall p \ v_0 \ s_0 \ t_0 \ r_1 \ r_2 \ b_1 \ b_2 \ v_1 \ v_2,$
 $\text{runT } p \ v_0 \ s_0 \ t_0 \ r_1 \ b_1 \ v_1 \rightarrow \text{runT } p \ v_0 \ s_0 \ t_0 \ r_2 \ b_2 \ v_2 \rightarrow r_2 = r_1 \wedge v_2 = v_1 \wedge b_2 = b_1. \quad (6)$

250 The lemma above tells that *runT* behaves deterministically. The proof is by induction
 251 on the first *runT* execution and the proof is rather immediate, since all rules are in exclusive
 252 due to the hypothesis on the *path_end*: the hypotheses *success* t_0 , *path_atom* t_0 and *failed* t_0
 253 are exclusive. Moreover, by Lemma 6, we have that the hypothesis of FAILT implies *failed* t_0 ,
 254 therefore it is exclusive wrt STOPT and STEPT, but also from BACKT since they have
 255 different expectations on the output of *next_alt*.

$$\begin{array}{c}
 \frac{\text{success } t_0 \quad \text{get_} s_0 \ t_0 = s_1 \quad \text{next_} \top t_0 = t_1}{\text{runT } v_0 \ s_0 \ t_0 \ \bullet(s_1, t_1) \perp v_0} \text{STOPT} \\
 \\
 \frac{\text{path_} t_0 \quad b_2 = \text{is_tg} \parallel b_1 \quad \text{step } p \ v_0 \ s_0 \ t_0 = (v_1, \text{tg}, t_1) \quad \text{runT } v_1 \ s_0 \ t_1 \ r \ b_1 \ v_2}{\text{runT } v_0 \ s_0 \ t_0 \ r \ b_2 \ v_2} \text{STEPT} \\
 \\
 \frac{\text{failed } t_0 \quad \text{next_} \perp t_0 = \bullet t_1 \quad \text{runT } v_0 \ s_0 \ t_1 \ r \ n \ v_1}{\text{runT } v_0 \ s_0 \ t_0 \ r \ n \ v_1} \text{BACKT} \\
 \\
 \frac{\text{next_} \perp t_0 = \square}{\text{runT } v_0 \ s_0 \ t_0 \ \square \perp v_0} \text{FAILT}
 \end{array}$$

■ **Figure 7** Rule system for *runT*

256 3.4 Properties of *runT*

257 Working with *runT* allows to prove some interesting properties about the behavior of the
 258 cut operator wrt disjunction, i.e. the *Or* node. We have proven several properties that are
 259 available at [this link](#). Below we show two of these lemmas.

260 **Lemma** *runSST_or*: $\forall p \ v_0 \ v_1 \ s_0 \ s_1 \ t_0 \ t'_0 \ \text{sm } t_1,$
 $\text{runT } p \ v_0 \ s_0 \ t_0 \ \bullet(s_1, \bullet t'_0) \top v_1 \rightarrow$
 $\text{runT } p \ v_0 \ s_0 \ (\text{Or } \bullet t_0 \ \text{sm } t_1) \ \bullet(s_1, \bullet(\text{Or } \bullet t'_0 \ \text{sm } \text{KO})) \perp v_1. \quad (7)$

261 **Lemma** *runNT_or*: $\forall p \ v_0 \ v_1 \ s_0 \ t_0 \ \text{sm } t_1,$
 $\text{runT } p \ v_0 \ s_0 \ t_0 \ \square \top v_1 \rightarrow$
 $\text{runT } p \ v_0 \ s_0 \ (\text{Or } \bullet t_0 \ \text{sm } t_1) \ \square \perp v_1. \quad (8)$

262 **Lemma** *run_orSST* $p \ v_0 \ v_2 \ s_0 \ s_1 \ \text{sm } t_0 \ t'_0 \ t_1 \ t'_1:$
 $\text{runT } p \ v_0 \ s_0 \ (\text{Or } \bullet t_0 \ \text{sm } t_1) \ \bullet(s_1, \bullet(\text{Or } \bullet t'_0 \ \text{sm } t'_1)) \perp v_2 \rightarrow$
 $\exists b, \text{runT } p \ v_0 \ s_0 \ t_0 \ \bullet(s_1, \bullet t'_0) \ b \ v_2 \wedge t'_1 = \text{if } b \text{ then } \text{KO} \text{ else } t_1. \quad (9)$

263 In Lemma 7, we evaluate the tree t_0 . This evaluation produces a success with signature σ_1
 264 and a new tree t'_0 . During the evaluation, a superficial cut in t_1 is traversed. This implies that
 265 the evaluation of $\bullet t_0 \vee_{s_m} t_1$ returns σ_1 as substitution, and the alternatives are $\bullet t'_0 \vee_{s_m} \text{KO}$.
 266 The right-hand side of the *Or* node is replaced with *KO* by $!_r$.

267 In Lemma 8, we are in a similar scenario: t_0 evaluates a superficial cut but, this time,
 268 produces a failure. This implies that the evaluation of $\bullet t_0 \vee_{s_m} t_1$ also fails.

Definition $\text{stepE } v_0 \text{ t s a gl} :=$
 $\text{let } (v_1, \text{rs}) := \text{bc p } v_0 \text{ t s in}$
 $(v_1, \text{save_alts a gl rs}).$

$$\frac{}{\text{runE } v_0 ((s_0, \emptyset) :: a) \bullet (s_0, a)} \text{STOPE}$$

$$\frac{\text{runE } v_0 ((s_0, g) :: \text{ca}) \text{ r}}{\text{runE } v_0 ((s_0, (\text{cut}, \text{ca}) :: g) :: _) \text{ r}} \text{CUTE}$$

$$\frac{\text{stepE } v_0 \text{ t } s_0 \text{ a g} = (v_1, \text{bs}) \quad \text{runE } v_1 (\text{bs} ++ a) \text{ r}}{\text{runE } v_0 ((s_0, (\text{call } t, _) :: g) :: a) \text{ r}} \text{CALLE}$$

$$\frac{}{\text{runE } _ \emptyset \square} \text{FAILE}$$

■ **Figure 8** Rule system for runE

Finally, in Lemma 9, we evaluate $\bullet t_0 \vee_{s_m} t_1$. The evaluation succeeds, producing the result $\bullet(s_1, \bullet t'_0 \vee_{s_m} t'_1)$. This means that t_0 still has t'_0 as unexplored alternatives. We deduce that the evaluation of t_0 produces the substitution σ_1 with alternatives t'_0 . However, we have no information about whether a superficial cut has been crossed in t_0 . If this is the case, we know that t'_1 is cut away by $!_r$; otherwise, t'_1 has not been explored and is equal to t_1 .

From these lemmas, it is clear how the order in which alternatives are evaluated changes the behavior of the program.

4 Stack semantics

We now introduce the ELPI semantics. The interpreter is close to that of the ELPI language and provides an operational semantics that is similar to the one proposed by Pusch in [16], which is in turn closely related to the semantics given by Debray and Mishra in [6, Section 4.3]. Pusch mechanized this semantics in Isabelle/HOL, together with several optimizations that are also present in the Warren Abstract Machine [20, 1].

We represent a state of the ELPI language thanks to the two mutual inductives shown below.

Inductive $\text{alts} := \text{nilA} \mid \text{consA of } (\Sigma * \text{goals}) \ \& \ \text{alts}$
with $\text{goals} := \text{nilG} \mid \text{consG of } (A * \text{alts}) \ \& \ \text{goals} .$

An ELPI state is an enhanced two-dimensional list. The outermost list represents a list of alternatives in disjunction, each accompanied by the substitution that should be used for their interpretation. The innermost list is a list of atoms, that is, the list of goals in conjunction. These goals are decorated with a pointer to an alternative list, which is used to keep track of where to jump when a cut is interpreted. We call these special alternatives the *cut-to* alternatives.

The ELPI interpreter is called runE and has the following type.

Inductive $\text{runE} : \mathcal{F}_v \rightarrow \text{alts} \rightarrow \text{option } (\Sigma * \text{alts}) \rightarrow \text{Prop} :=$ (10)

The inductive runE represents a routine taking as input a set of variables and a list of alternatives and returning an option. $\square$ stands for interpretation with no solution, whereas

294 •(σ, a) represent a successful interpretation with σ as solution and a as the list of non-yet-
 295 explored choice points.

296 The rule system for *runE* is given in Figure 8 and is made by 4 cases. We distinguish two
 297 main cases: if the list of alternatives is empty, we have a failing state and need to return and
 298 two are the main scenarion. If the list of alternatives is empty, (FAILE) we have a failure:
 299 we stop the computation and return $\square$. If the list of alternatives is the list (gs, a), we again
 300 distinguish two main cases. If gs is

TODO:backE
 does not exist
 anymore

301 If the list of alternatives is non empty, we need to consider its

302 Otherwise, different cases depending on the shape of the alternatives. (STOPE) consider
 303 a non empty list of alternatives where the list of goals is empty. In this case, we have, by
 304 definition a success and return the input substitution and the tail of alternative. (CUTE)
 305 treats the cut operator, the

306 The idea of the ELPI interpreter is to receive a list of alternatives. The first alternative
 307 consists of a list of current goals. Four cases must be taken into account; they are shown in
 308 Figure 8. In order to simplify goal retrieval, we split the head of the alternatives from the
 309 tail, so that it can be immediately matched in the inductive definition. Note that an empty
 310 list of alternatives represents, by definition, a failing elpi state. If the goal list is empty
 311 (STOPE), then we have, by definition, a success, and the input solution together with the
 312 list of alternatives is returned. If the goal list starts with a cut (CUTE), then the current
 313 alternatives are erased in favour of the cut alternatives, and a recursive call is made on the
 314 remaining goal list.

315 Finally, we must consider the case in which the goal list starts with a call. The call
 316 can either fail (FAILE) or succeed (CALLE). We distinguish the two cases by looking if the
 317 backchaining operation returns zero or more rules. We have wrapped this task in the *stepE*
 318 procedure, which also updates the goal and cut-alternative list. The fail case, is relatively
 319 easy: the first goal does not succeed, we need to take the head of the alternatives, and make
 320 it the new list of goals to be explored.

321 The case in which backchaining produces a non empty list, the *save_alts* routine is in
 322 cahrg of: taking the list of premises and add to each atom the the list of alternatives a as
 323 their new cut-alternatives, then it append the list of goals gl to each of these new lists.

324 *Remark* It is not possible to prove the same lemma as in section xxx, because no structure,
 325 the scope of the cut ... spiegare bene il problema so that i can reuse this subsection has a
 326 hook when I will say that l2t is difficult + Yves

327 5 Equivalence of the two semantics

328 The equivalence between the two semantics is possible under two conditions: we need to
 329 work with “valid tree”, i.e. all of those trees that can be generated from a call. Secondly,
 330 we need to translate trees into elpi states. In the next to subsection we propose the two
 331 functions *valid_state* and *t2l*.

332 5.1 Valid trees

333 The inductive tree allows one to generate a large number of trees, some of which are not
 334 valid, in the sense that they cannot be produced starting from a given query. The class of
 335 valid trees is characterized by the function shown in Figure 9a.

336 While all base cases of tree are considered valid, we need to analyze carefully the cases
 337 for the *Or* and *And* constructors.

```

Fixpoint t2l A s0 bt :=
  match A with
  | OK          => [:: (s0, \mnil)]
  | KO          => \mnil
  | TA a        => [:: (s0, [:: (a, \mnil) ])]
  | Or □ s1 B => add_ca_deep bt (t2l B s1 \mnil)
  | Or •A s1 B  =>
    let lB := t2l B s1 \mnil in
    let lA := t2l A s0 lB in
    add_ca_deep bt (lA \mappend lB)
  | And A B0 B  =>
    let lA := t2l A s0 bt in
    let lB0 := a2g B0 in
    let lA := add_deep bt lB0 lA in
    if lA is [:: (s0, gs) & al] then
      let al := map (catr lB0) al in
      let lB := t2l B s0 (al \mappend bt) in
      map (catl gs) lB \mappend al
    else \mnil
  end.

Fixpoint valid_tree A :=
  match A with
  | TA _ | OK | KO => T
  | Or □ _ B => valid_tree B
  | Or •A _ B => valid_tree A &&
    ((B == KO) || B.base_or B)
  | And A B0 B => valid_tree A &&
    if success A then valid_tree B
    else B == big_and B0
  end.

```

(a) Valid tree

(b) Tree to list

■ **Figure 9** Valid tree and Tree to list

338 For the *Or* constructor, we distinguish two cases depending on whether the left-hand
 339 side (lhs) exists. If it does not exist, then the right-hand side (rhs) must be a valid tree.
 340 Otherwise, the lhs must itself be a valid tree, and the rhs is either the *KO* tree, since it may
 341 have been removed by the evaluation of a superficial cut in the lhs, or it has not yet been
 342 explored. In the latter case, it is a *base_or* tree, namely the right-skewed tree formed by a
 343 disjunction of conjunctions that is generated after a successful backchain to a call.

344 For the *And* constructor, the lhs is required to be a valid tree. The shape of the rhs
 345 depends on whether the lhs is a success. If the lhs is not successful, then the rhs has never
 346 been explored: the procedures *step* and *next_alt* modify the rhs only when the lhs succeeds.
 347 In this case, the lhs must be the right-skewed tree containing conjunctions of premises atoms,
 348 the reset point B_0 ensure what shape the rhs should have. If the lhs is a success tree, then
 349 the rhs can be modified by *step* and *next_alt*, therefore it must be a valid tree.

350 5.2 From trees to lists

351 The translation of a tree to a list is shown in Figure 9b. It takes the tree to be translated, a
 352 substitution (called s), a list of alternatives, called bt . The substitution s tells what is the
 353 substitution of the alternative we are building and is updated when going to the rhs of the
 354 *Or* constructor. The bt list represents the future alternatives list that have the same depth
 355 of the current tree root. They are useful to know if they should be added to the current goal
 356 as cut alternatives.

357 An *OK* node represent a success in a tree, then the translation of this tree returns a
 358 singleton list with the couple $(s, [::])$: there is one alternative with 0 goals, i.e. a success in
 359 elpi by STOPE. The *KO* node represent a failure, and, therefore 0 alternatives. An atom is
 360 translated into a singleton with the atom as first goal and the empty cut-alternative list: we
 361 are not adding bt since they are the alternatives for the right-sibling, not the right-uncles,

362 this means that if we have a is a cut, then it erases the alternatives bt .

363 In a disjunction, we are scendendo di un livello the distance from our backtracking list.
 364 This means that bt becomes the list of right-uncles of the two branches of the Or constructor.
 365 The compilation of the rhs is done independently of if the lhs exists: we transform the rhs
 366 into an elpi state and passing the empty list of alternatives, since the rhs as no right-siblings.
 367 Then, thanks to `add_ca_deep`, we recursively add bt to every cut-alternative appearing in
 368 the translated goals. If the lhs exists, we translate it and pass the translation of the rhs as
 369 the list of right-siblings. Then we prepend the translated lhs to the translated rhs and add
 370 bt with `add_ca_deep`.

371 We compile the *And* case by translating its lhs to the list lA and reset point to the list
 372 $lB0$. Then, thanks to `add_deep`, we recursively append $lB0$ to the alternatives born in lA ,
 373 i.e. we leave unchanged the pointers to bt , if any, in lA . We finally need to treat cases
 374 whenever the list lA is empty or not. In the former case the empty list is returned, otherwise,
 375 we have a list $[:: (s_0, gs) \& al]$. By the semantics of the *And* in the *runT* interpreter, the rhs
 376 of the *And* represent the sequent of goals to be executed after the first alternative in the
 alternative in the lhs, it corresponds to the list of goals gs . The reset point $B0$ is to be appended to the next
 tree? rephrase 378 alternatives in the lhs, that is the queue of lA , that is al .

379 We note therefore that the compilation of lB is done by passing the alternatives $al + +bt$
 380 since the alternatives in al are to be...

381 5.3 Equivalence theorems

382 We have now all the ingredients that ...

```

383 Lemma tree_to_elpi fv A s1 b fv' r :
      vars_tree A `<= ` fv → vars_sigma s1 `<= ` fv →
      valid_tree A →
      runT u p fv s1 A r b fv' →
      let xs := t2l A s1 \mnil in
      let r' := omap (fun '(s, t) => (s, (t2l (odflt KO t) s1 \mnil))) r in
      runE u p fv xs r'.
      (10)

```

```

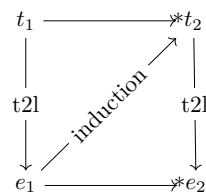
Lemma elpi_to_tree v0 a r :
      runE u p v0 a r →
      ∀ s0 t, valid_tree t → t2l t s0 \mnil = a →
      ∃ b v1,
      if r is •(s1, a') then
        ∃ t1, runT u p v0 s0 t •(s1, t1) b v1 ∧ t2l (odflt KO t1) s0 \mnil = a'
      else runT u p v0 s0 t □ b v1.
      (11)

```

385 The proof of Lemma 11 is based on the idea explained in [2, Section 3.3]. An ideal
 386 statement for this lemma would be: given a function $l2t$ transforming an elpi state to a
 387 tree, we would have have that the the execution of an elpi state e is the same as executing
 388 *runT* on the tree resulting from $l2t(e)$. However, it is difficult to retrieve the strucutre of
 389 an elpi state and create a tree from it. This is because, in an elpi state, we have no clear
 390 information about the scope of an atom inside the list and, therefore, no evident clue about
 391 where this atom should be place in the tree.

392 Our theorem states that, starting from a valid tree t which translates to a list of alternatives
 393 $(\sigma_1, g) :: a$. If we run in elpi the list of alternatives, then the execution of the tree t returns the
 394 same result as the execution in elpi. The proof is performed by induction on the derivations
 395 of the elpi execution. We have 4 derivations.

396 We have 4 case to analyse:



■ **Figure 10** Induction scheme for Lemma 11

6 Case study: determinacy analysis

we mechanize the first order part of xxx.

snippet det, main thm, invariant det tree (valid tree prev section?)

proof induction on exec, step/next alt preserving invariant proved by induction on the tree.

with list semantics cut and next alt requires to express a link between the ca or next alts and the current goal, which is non trivial without an intermediate data structure like the tree

7 Related work

The only mechanization of Prolog's semantics we could find is the one by Pusch in ISABELLE/HOL [16] that is based on the model of Debray and Mishra [6, Sec. 4.3]. Their work start from that semantics and refines it by introducing some optimizations usually found in the Warren abstract machine [20, 1]. They do not use the semantics to prove properties on the execution of programs as we do in our use case, hence they do not face the difficulty described in section XXX that pushed us to develop an more explicit semantics (with respect to cut). In some way our work is complementary, showing the stack based semantics equivalent to more optimized one, we could consider implement in Elpi.

Another relevant work are the ones by King cite, but we could not find their code. Their semantics is denotational and cannot easily cover all the features our paper XX does, but would have been sufficient for this first step.

The proof technique we use to relate the two semantics is inspired by XXX and we thank Y.Bertot for insightful discussions on the topic. In XXX Bertot relates bla bla, without never constructing a decompiler, exactly as we never build a list to tree function.

8 Conclusion

We do this. This is a first step for XXXX. We need to add substitutions. The missing part of the proof becomes related to abstract interpretation, since the det types for a lattice and one has to prove that the concrete runtime values variables take agree with the abstraction computed statically. This proof is ongoing and somehow orthogonal to the current one since the cut operator and the HO feature do not interfere with each other in complex ways.

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